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SHORT-PAPER

Enhancing Game Mechanics for IP-Based Games: The IP-MDA Framework with a Case Study on the White Day Korean Horror IP

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Enhancing Game Mechanics for IP-Based Games

The IP-MDA Framework with a Case Study on the White Day Korean Horror IP

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Abstract

This paper introduces the IP-MDA framework, a modified version of the Mechanics-Dynamics-Aesthetics (MDA) framework designed to facilitate the production of original games based on intellectual property (IP). By incorporating established transmedia world (TMW) elements such as mythos, ethos, and topos, the IP-MDA framework enhances traditional game design principles. Using the popular Korean horror game series White Day as a case study, we demonstrate how the IP-MDA framework can address the challenges of integrating external IP into game mechanics and dynamics, while preserving the aesthetic goals of the original work. Through several iterations, the framework was tested and validated to improve how game mechanics are shaped to reflect and leverage the underlying IP.

CCS Concepts

• General and reference → Design;

Keywords

IP-MDA, MDA framework, intellectual property, game design, mechanics-dynamics-aesthetics

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1 Introduction

1.1 Background of Study

Games that incorporate intellectual properties (IPs) have become a strategic approach in the gaming industry, often raising fan anticipation and generating significant added value[9]. Various transmedia strategies and media franchise approaches have been extensively researched[2, 3, 9, 12], but these studies have predominantly focused on narrative elements, overlooking unique aspects of games such as interactivity[13] and stories generated through player presence. This project involved creating a visual novel game by adapting an existing IP while preserving TMW elements. This study represents an attempt to modify the widely adopted MDA framework[7] to enable the creation of games utilizing IPs[5].

1.2 IP-MDA Framework

The IP-MDA framework retains the original MDA structure while incorporating modifications to account for the universe of the source IP and the elements derived from it for the game IP. The source IP's universe is analyzed through the transmedia world core elements—Mythos, Ethos, and Topos[8]—and this analysis is employed to structure the foundation of the game IP using the five dramatic elements of games: challenge, story, play, character, and premise[6].

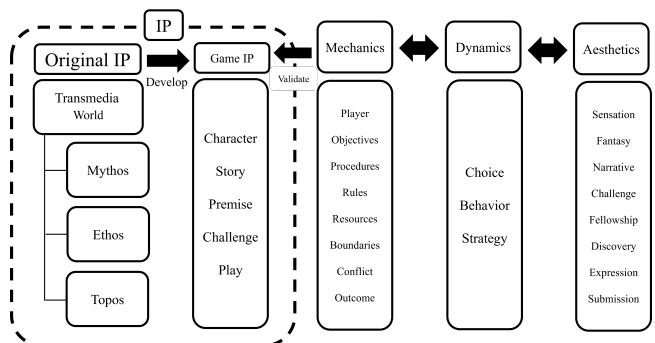


Figure 1: IP-MDA Framework

2 Application of the IP-MDA Framework: White Day Game Project

2.1 White Day Game Overview

The White Day game series, originally developed by Sonnori[11], revolves around the "strange events that occur at Yeondu High School." The game garnered significant popularity in South Korea and gained international recognition through its Steam release. The combination of engaging storylines and well-developed characters has earned the game a loyal fanbase.

2.2 Analysis of the Transmedia World

Based on the framework of transmedia world(TMW)[8] core elements, we analyzed:

Mythos (Core Knowledge for Understanding the Story World):

- Male protagonist with romantic feelings for heroines
- Characters including ghosts based on Korean folklore
- School's barrier system based on Yin-Yang and the Five elements

Topos (A Specific Point in History and Geopolitical, Physical Space):

- Events occurring around White Day(14th March)
- Yeondu High School setting
- Timeline from March 13-15, 2001

Ethos (Embedded or Expressed Ethical Norms):

- Korean cultural ethical norms
- Ghost-related threats and possession
- Themes of revenge and unresolved grudges

2.3 Game IP Development Through TMW Analysis

Based on the analysis of White Day's TMW, we established settings for our game using the Game IP component of the IP-MDA framework[5]. The five dramatic elements were either closely aligned with the original or adapted from key components considered essential, maintaining continuity with the IP.

In order to proceed, we aimed to define the criteria for developing these five elements by referencing both the theoretical analysis of the TMW and the definitions of Fullerton's five dramatic elements[6].

Character: Characters are the "beings" that move within the story, representing the player or existing in relationships with the player. Following the original IP's approach, we matched the number of heroines and recreated their traits to align with the core keywords of White Day's TMW, such as Yin-Yang and the Five Elements. The protagonist was endowed with traits of bravery and responsibility.

Story: The story governs the overall content and provides background for the game's primary objectives[6]. The basic plot follows the protagonist as he works to rescue the heroine trapped in the school and escape together, adhering to the original IP's storyline while modifying the temporal setting to better align with contemporary players' perceptions.

Premise: The premise serves as a device for players to naturally recognize foundational situations and actions[6]. We incorporated the normative framework drawing upon Korean elements of the original world-building, including the cultural sentiment of Han associated with Korean ghosts.

Challenge: Challenges represent the goals that players must achieve, constructed based on the Mythos background narrative and interaction knowledge. The original IP challenged players to avoid ghosts and the janitor, solve puzzles, and escape from the school. We maintained these elements while ensuring appropriate difficulty progression.

Play: Play aids in player immersion and creates situations for free action within the game's structure. Players can gather various clues, solve puzzles, and make strategic choices, maintaining the original IP's engagement patterns.

2.4 Iteration Process of Applying the IP-MDA Framework to the White Day Project

After conducting the TMW analysis of the IP and developing the Game IP, we aimed to apply the MDA process and verification methodology of the IP-MDA framework to the game's main system development. During this process, the game IP served as both the objective and the subject of verification. To validate this approach, the development team aimed to verify the main system using the five dramatic elements proposed by Tracy Fullerton[6], which align with the five elements of the Game IP. We sought to ensure that the mechanics reflected the key characteristics of the IP and that these mechanics would allow players to experience the intended aesthetics. Based on the analysis of each element, we derived the following framework for a basic checklist.

- Can the (Mechanics) be established based on the (Thesis-TMW, five dramatic elements)?

This question ultimately needed to align with our goal of "rescuing the heroine" while also verifying whether the mechanics reflected the established Game IP. Therefore, we aimed to formulate specific questions based on the structured questions that incorporated the actual content of the Game IP. Additionally, since the main system had to be one that specifically guided the actions of the "playable character," the questions were further refined with a focus on the protagonist, Seo Ji-tae. The questions were modified to fit each mechanic, and iterations were conducted accordingly. The checklist was tested by members of the development team, with the roles of each team member defined as follows. Each team member independently completed the checklist, and after completing the process, we conducted interviews to discuss any areas that needed improvement.

Table 1: Participant Numbers and Roles in the Verification Process

No.	Role
P1	Director
P2	Game Designer1
P3	Game Designer2
P4	UI Designer
P5	Scenario Writer
P6	Programmer
P7	Art Designer

The iterations were conducted as follows:

First iteration

- Aesthetics : perceived to eliminate evil ghost
- Dynamics : use tools, killing
- Mechanics : rules for eliminating, can use tools

In the first attempt, we set the aesthetic goal of making the player perceive the "evil ghost as entities that must be eliminated" by utilizing the existing attitudes and methods towards evil ghost found in White Day. To achieve this, we analyzed the behavior of the ghosts, particularly those that held some of the heroines captive, from the original White Day system. We also examined the commands and methods used to confront these evil ghost. However, directly applying this approach posed challenges due to the genre-specific characteristics of visual novels, which focus on reading and branching story choices. This difference made it difficult to integrate the original system. We identified this as the first issue to address, and to resolve it, we introduced a mechanic where the tools to defeat evil ghost are presented as scripts and choices.

Subsequently, we created three questions per dramatic element, reflecting the mechanics, and conducted verification with seven developers[Figure 2]. Based on this verification, we concluded that simply using tools to defeat ghosts and rescue the heroines does not sufficiently convey Seo Ji-tae's moral actions, nor does it effectively reflect the han sentiment associated with Korean ghosts. Additionally, this approach alone does not adequately highlight Seo Ji-tae's role. Thus, we recognized the need to emphasize his role as a "shaman" in order to better express the Han sentiment.

Mechanics: Seo Ji-tae uses a tool (presented as a choice) to defeat the ghost.								
Thesis	Dramatic Element	Checklist	P1	P2	P3	P4	P5	P7
Ethos	Character	Is Seo Ji-tae a character capable of defeating ghosts using tools?	Yes	Yes	Yes	Yes	Yes	Yes
Ethos	Character	Is Seo Ji-tae's act of defeating ghosts with tools driven by good intentions?	Yes	No	No	No	Yes	No
Topos	Character	Does defeating ghosts with tools make both ghosts and characters exist in the given background, Yeondu High School?	Yes	Yes	Yes	Yes	Yes	Yes
Story	Story	Can the act of defeating ghosts using tools enable the game to progress with the storyline of rescuing the heroines and escaping?	Yes	Yes	Yes	No	Yes	Yes
Topos	Story	Can the act of Seo Ji-tae defeating ghosts with tools reflect the storyline set in the current backdrop of 'Yeondu High School'?	Yes	No	Yes	Yes	Yes	Yes
Ethos	Story	Can the storyline unfold with Seo Ji-tae acting out of good intentions and the ghosts acting out of malicious intent, through Seo Ji-tae's act of defeating ghosts with tools?	Yes	Yes	No	Yes	Yes	Yes
Mythos	Premise	Does Seo Ji-tae's act of defeating ghosts with tools utilize shamanism, which is part of the game's established background knowledge of IP?	Yes	Yes	Yes	Yes	Yes	Yes
Mythos	Premise	Does Seo Ji-tae's act of defeating ghosts with tools reflect the han sentiment, a characteristic of Korean culture?	Yes	No	Yes	No	Yes	Yes
Ethos	Premise	Is Seo Ji-tae's act of exorcising ghosts using a 'truth item' that corresponds to the reason for their death designed to function within the opposing dynamics of good versus evil?	No	Yes	No	Yes	Yes	No
Challenge	Challenge	Can Seo Ji-tae's act of defeating ghosts with tools serve as a gimmick that the he must solve, as a protagonist trapped in the school, in order to escape?	Yes	Yes	Yes	Yes	Yes	Yes
Challenge	Challenge	Can Seo Ji-tae's act of defeating ghosts with tools be a recurring action within the game?	Yes	Yes	Yes	Yes	Yes	Yes
Challenge	Challenge	Can Seo Ji-tae's act of defeating ghosts with tools create variations in difficulty levels, either making the task harder or easier?	Yes	Yes	Yes	Yes	No	Yes
Play	Play	Can Seo Ji-tae's act of defeating ghosts with tools provide the player with a fun of thrill?	Yes	No	Yes	No	Yes	Yes
Play	Play	Can Seo Ji-tae's act of defeating ghosts with tools provide the player with a fun of storytelling?	Yes	Yes	Yes	No	No	Yes
Play	Play	Can Seo Ji-tae's act of defeating ghosts with tools affect the outcome or alter the final result screen for the player?	Yes	Yes	Yes	Yes	Yes	Yes

Figure 2: First Iteration's Verification Result**Second iteration**

- Aesthetics : understand the truth of evil ghost death
- Dynamics : find the truth, use truth item
- Mechanics : exorcise the ghost, can use truth item

To address the issues from the first attempt, we explored a mechanic that would allow the Han sentiment of the evil ghost to be conveyed, focusing on the question, "Why did that guy become a ghost?" This led us to consider a mechanic in which the player's aesthetic goal would be to "understand the truth behind that ghost."

The truth behind the ghosts, specifically the reason for its death, was designed to be revealed through clues, a common dynamic found in visual novels and various games. This mechanic would guide the player through uncovering the truth, thereby driving the player's actions.

With this approach, we designed a mechanic where the player presents the truth to the ghost, offering an item that corresponds to the truth, thereby revealing the reason for the ghost's death. This would allow the ghost to find peace (exorcism), progressing the story in a way that reflects the Han sentiment.

Upon analyzing the verification results[Figure 3], the mechanics from the second iteration partially addressed the limitation of the first iteration, namely the lack of the emotional depth of Han. However, some members of the development team still raised doubts about Seo Ji-tae's moral alignment. Additionally, it was argued that using the "truth item" to reveal the cause of the ghost's death could result in the ghosts losing their inherently malicious nature.

Mechanics: Seo Ji-tae exorcises the ghosts by using a 'truth item' that corresponds to the reason for their death.											
Thesis	Dramatic Element	Checklist	P1	P2	P3	P4	P5	P6	P7		
Ethos	Character	Can the character Seo Ji-tae perform the act of exorcising ghosts by using a 'truth item' that corresponds to the reason for their death?	Yes	Yes	Yes	Yes	Yes	Yes	Yes		
Ethos	Character	Is Seo Ji-tae's act of exorcising ghosts using a 'truth item' that corresponds to the reason for their death an action driven by good intentions?	Yes	No	No	No	Yes	No	Yes		
Topos	Character	Does exorcising ghosts using a 'truth item' that corresponds to the reason for their death make both ghosts and characters exist in the given background, Yeondu High School?	Yes	Yes	Yes	Yes	Yes	Yes	Yes		
Story	Story	Can the act of Seo Ji-tae exorcising the ghosts using a 'truth item' that corresponds to the reason for their death enable the game to progress with the storyline of rescuing the heroines and escaping?	Yes	Yes	Yes	No	Yes	Yes	Yes		
Topos	Story	Can the act of Seo Ji-tae exorcising ghosts using a 'truth item' that corresponds to the reason for their death reflect the storyline set in the current backdrop of Yeondu High School?	Yes	No	Yes	Yes	Yes	Yes	Yes		
Ethos	Story	Can the storyline unfold with Seo Ji-tae acting out of good intentions and the ghosts acting out of malicious intent, through Seo Ji-tae exorcising ghosts using a 'truth item' that corresponds to the reason for their death?	Yes	Yes	No	Yes	No	No	Yes		
Mythos	Premise	Does Seo Ji-tae's act of exorcising ghosts using a 'truth item' that corresponds to the reason for their death reflect the han sentiment, to the reason for their death utilize shamanism, which is part of the game's established knowledge of IP?	Yes	Yes	Yes	Yes	Yes	Yes	Yes		
Mythos	Premise	Does Seo Ji-tae's act of exorcising ghosts using a 'truth item' that corresponds to the reason for their death reflect the han sentiment, a characteristic of Korean culture?	Yes	No	Yes	No	Yes	Yes	No		
Ethos	Premise	Is Seo Ji-tae's act of exorcising ghosts using a 'truth item' that corresponds to the reason for their death designed to function within the opposing dynamics of good versus evil?	No	Yes	No	Yes	Yes	No	Yes		
Challenge	Challenge	Can Seo Ji-tae's act of exorcising ghosts using a 'truth item' serve as a gimmick that the he must solve, as a protagonist trapped in the school, in order to escape?	Yes	Yes	Yes	Yes	Yes	Yes	Yes		
Challenge	Challenge	Can Seo Ji-tae's act of exorcising ghosts using a 'truth item' that corresponds to the reason for their death be a recurring action within the game?	Yes	Yes	Yes	Yes	Yes	Yes	Yes		
Challenge	Challenge	Can Seo Ji-tae's act of exorcising ghosts using a 'truth item' that corresponds to the reason for their death create variations in difficulty, making the task harder or easier?	Yes	No	Yes	Yes	Yes	No	Yes		
Play	Play	Can Seo Ji-tae's act of exorcising ghosts using a 'truth item' that corresponds to the reason for their death provide the player with a fun of thrill?	No	No	Yes	No	No	No	No		
Play	Play	Can Seo Ji-tae's act of exorcising ghosts using a 'truth item' that corresponds to the reason for their death provide the player with a fun of storytelling?	Yes	Yes	Yes	Yes	Yes	Yes	Yes		
Play	Play	Can Seo Ji-tae's act of exorcising ghosts using a 'truth item' that corresponds to the reason for their death affect the final result screen for the player?	Yes	Yes	Yes	Yes	Yes	Yes	Yes		

Figure 3: Second Iteration's Verification Result**Third iteration**

- Aesthetics : understand the truth of evil ghost death
- Dynamics : find the truth, distinguishing between truth and false
- Mechanics : talking with ghost and forming rapport

In the second attempt, while we established a mechanic centered around uncovering the truth regarding the Han sentiment of the evil ghost, it remained challenging to make this easily comprehensible for the player at a glance. Therefore, there was a need for a method

that would allow players to quickly grasp the truth about the ghosts. Additionally, this method needed to create a sense of thrill and difficulty, necessitating the development of specific mechanics.

To address this, we decided to categorize the truths using the 5W1H, clarifying the boundary between truth and falsehood. We aimed to apply this method of mechanics to fit the visual novel format.

However, there was a concern that this approach might be limited to merely finding the truth. Moreover, it was essential to overcome the previous iteration's limitation of not sufficiently drawing out Seo Ji-tae's good qualities while still emphasizing the Han sentiment. Thus, we sought to incorporate the concept of "rapport," where Seo Ji-tae and the ghosts build a connection to help recover their memories. Rapport is generally understood as a positive emotional connection that occurs between a counselor and a client[4, 14], which can be described as an authentic interaction, a favorable emotional response between individuals[1], and a supportive and empathetic relationship [10].

Verification[Figure 4] results indicated that while the 5W1H mechanics effectively engaged the development team in creating challenges, the rapport concept diminished the ghosts' malevolence. The ghosts were perceived more as wronged souls than malicious entities, undermining the IP's intended evil attributes. However, the team agreed on the importance of mechanics that emphasize Seo Ji-tae's good intentions through memory recovery.

Mechanics: Seo Ji-tae utilizes the 5W1H to uncover the reasons behind the ghosts' deaths while previously forming rapport to retrieve their memories.									
Thesis	Dramatic Element	Checklist	P1	P2	P3	P4	P5	P6	P7
Ethos	Character	Can the character Seo Ji-tae utilize the 5W1H to uncover the reasons behind the ghosts' deaths while forming rapport to retrieve their memories?	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ethos	Character	Is Seo Ji-tae's act of utilizing the 5W1H to uncover the reasons behind the ghosts' deaths while forming rapport to retrieve their memories an action driven by good intentions?	Yes	Yes	No	Yes	Yes	No	Yes
Topos	Character	Does Seo Ji-tae's act of utilizing the 5W1H to uncover the reasons behind the ghosts' deaths while forming rapport to retrieve their memories make both ghosts and characters exist in the given background, Yeondu High School?	Yes	Yes	Yes	Yes	No	Yes	Yes
Story	Story	Can the Seo Ji-tae's act of utilizing the 5W1H to uncover the reasons behind the ghosts' deaths while forming rapport to retrieve their memories enable the game to progress with the storyline of rescuing the heroines and escaping?	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Topos	Story	Can Seo Ji-tae's act of utilizing the 5W1H to uncover the reasons behind the ghosts' deaths while forming rapport to retrieve their memories reflect the storyline set in the current backdrop of Yeondu High School?	Yes	Yes	Yes	Yes	No	Yes	Yes
Ethos	Story	Can the storyline unfold with Seo Ji-tae acting out of good intentions and the ghosts acting out of malicious intent through Seo Ji-tae's act of utilizing the 5W1H to uncover the reasons behind the ghosts' deaths while forming rapport to retrieve their memories?	No	No	No	No	No	No	Yes
Mythos	Premise	Does Seo Ji-tae's act of utilizing the 5W1H to uncover the reasons behind the ghosts' deaths while forming rapport to retrieve their memories utilize shamanism, which is part of the game's established knowledge of IP?	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mythos	Premise	Does Seo Ji-tae's act of utilizing the 5W1H to uncover the reasons behind the ghosts' deaths while forming rapport to retrieve their memories reflect the Han sentiment, a characteristic of Korean culture?	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ethos	Premise	Is Seo Ji-tae's act of utilizing the 5W1H to uncover the reasons behind the ghosts' deaths while forming rapport to retrieve their memories designed to function within the opposing dynamics of good versus evil?	No	No	No	No	No	No	No
Challenge	Challenge	Can Seo Ji-tae's act of utilizing the 5W1H to uncover the reasons behind the ghosts' deaths while forming rapport to retrieve their memories serve as a gimmick that the he must solve, as a protagonist trapped in the school, in order to escape?	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Challenge	Challenge	Can Seo Ji-tae's act of utilizing the 5W1H to uncover the reasons behind the ghosts' deaths while forming rapport to retrieve their memories be a recurring action within the game?	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Challenge	Challenge	Can Seo Ji-tae's act of utilizing the 5W1H to uncover the reasons behind the ghosts' deaths while forming rapport to retrieve their memories provide the player with a fun of thrill?	No	Yes	Yes	Yes	Yes	Yes	Yes
Play	Play	Can Seo Ji-tae's act of utilizing the 5W1H to uncover the reasons behind the ghosts' deaths while forming rapport to retrieve their memories create variations in difficulty, making the task harder or easier?	Yes	No	Yes	Yes	No	Yes	Yes
Play	Play	Can Seo Ji-tae's act of utilizing the 5W1H to uncover the reasons behind the ghosts' deaths while forming rapport to retrieve their memories provide the player with a fun of storytelling?	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Play	Play	Can Seo Ji-tae's act of utilizing the 5W1H to uncover the reasons behind the ghosts' deaths while forming rapport to retrieve their memories affect the final result screen for the player?	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Figure 4: Third Iteration's Verification Result

Fourth iteration

- Aesthetics : understand the truth of evil ghost death, experience the "evil"
- Dynamics : check the memory of the truth, avoding triggers
- Mechanics : avoid the ghost's Yeokrin, 5W1H

In the third attempt, while the understanding of the truth behind the evil ghosts and the retrieval of memories were addressed through the 5W1H, the entry point into the stage of applying the

5W1H concerning the remaining evil ghosts required further contemplation. Even with the sentiment of Han, the ghosts fundamentally represent malevolent entities, and transforming this into something with a "good" connotation did not align with the characteristics of the game IP we were developing.

For the manifestation of "evil," specific triggers are necessary. We contemplated how the players might either evade or confront these triggers to deepen their experience of the ghosts' malevolence, drawing inspiration from the term "Yeokrin(hit a nerve)". Yeokrin refers to a person's vulnerability that should not be touched. If this sensitive area is provoked, players would experience the evil of the ghosts, leading to a deeper understanding of the ghosts' nature and the circumstances of their deaths, ultimately allowing for the discovery of truths within the mechanics.

In the event of rescuing the heroine, players would engage in a dialogue with the ghosts, listening to their stories while avoiding the trigger (the Yeokrin), allowing the difficulty of the 5W1H mechanics to vary based on the level of malevolence exhibited by the ghosts. To enhance this difficulty, we introduced a time constraint as a device to create urgency and tension, which we believed would further complement the existing difficulty mechanics.

Through the fourth iteration's verification process, it was found that the Yeokrin concept successfully maintained the ghosts' malevolent nature while addressing the Han-related limitations. While concerns about diminished malevolence and temporal difficulty specificity were raised, the team determined these could be resolved through presentation and narrative enhancements. The final design involved Seo Ji-tae using the 5W1H system to uncover the ghosts' death reasons while avoiding their Yeokrin and identifying truths within time limits, effectively balancing all key elements.

Mechanics : Seo Ji-tae utilizes the 5W1H to uncover the reasons behind the ghosts' deaths while avoiding the ghosts' yeokrin and accurately identifying the truth within a set time limit.									
Thesis	Dramatic Element	Checklist	P1	P2	P3	P4	P5	P6	P7
Ethos	Character	Can the character Seo Ji-tae utilize the 5W1H to uncover the reasons behind the ghosts' deaths while avoiding the ghosts' yeokrin and accurately identifying the truth within a set time limit?	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ethos	Character	Is Seo Ji-tae's act of utilizing the 5W1H to uncover the reasons behind the ghosts' deaths while avoiding the ghosts' yeokrin and accurately identifying the truth within a set time limit an action of a virtuous individual?	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Topos	Character	Does Seo Ji-tae's act of utilizing the 5W1H to uncover the reasons behind the ghosts' deaths while avoiding the ghosts' yeokrin and accurately identifying the truth within a set time limit make both ghosts and characters exist in the given background, Yeondu High School?	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Story	Story	Can the Seo Ji-tae's act of utilizing the 5W1H to uncover the reasons behind the ghosts' deaths while avoiding the ghosts' yeokrin and accurately identifying the truth within a set time limit enable the game to progress with the storyline of rescuing the heroines and escaping?	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Topos	Story	Can Seo Ji-tae's act of utilizing the 5W1H to uncover the reasons behind the ghosts' deaths while avoiding the ghosts' yeokrin and accurately identifying the truth within a set time limit reflect the storyline set in the current backdrop of Yeondu High School?	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ethos	Story	Can the storyline unfold with Seo Ji-tae acting out of good intentions and the ghosts acting out of malicious intent through Seo Ji-tae's act of utilizing the 5W1H to uncover the reasons behind the ghosts' deaths while avoiding the ghosts' yeokrin and accurately identifying the truth within a set time limit?	Yes	Yes	Yes	No	Yes	Yes	Yes
Mythos	Premise	Does Seo Ji-tae's act of utilizing the 5W1H to uncover the reasons behind the ghosts' deaths while avoiding the ghosts' yeokrin and accurately identifying the truth within a set time limit utilize shamanism, which is part of the game's established knowledge of IP?	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mythos	Premise	Does Seo Ji-tae's act of utilizing the 5W1H to uncover the reasons behind the ghosts' deaths while avoiding the ghosts' yeokrin and accurately identifying the truth within a set time limit reflect the Han sentiment, a characteristic of Korean culture?	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ethos	Premise	Is Seo Ji-tae's act of utilizing the 5W1H to uncover the reasons behind the ghosts' deaths while forming rapport to retrieve their memories designed to function within the opposing dynamics of good versus evil?	Yes	Yes	Yes	Yes	Yes	No	Yes
Challenge	Challenge	Can Seo Ji-tae's act of utilizing the 5W1H to uncover the reasons behind the ghosts' deaths while avoiding the ghosts' yeokrin and accurately identifying the truth within a set time limit serve as a gimmick that the he must solve, as a protagonist trapped in the school, in order to escape?	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Challenge	Challenge	Can Seo Ji-tae's act of utilizing the 5W1H to uncover the reasons behind the ghosts' deaths while avoiding the ghosts' yeokrin and accurately identifying the truth within a set time limit be a recurring action within the game?	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Challenge	Challenge	Can Seo Ji-tae's act of utilizing the 5W1H to uncover the reasons behind the ghosts' deaths while avoiding the ghosts' yeokrin and accurately identifying the truth within a set time limit create variations in difficulty, making the task harder or easier?	Yes	Yes	Yes	Yes	Yes	No	Yes
Play	Play	Can Seo Ji-tae's act of utilizing the 5W1H to uncover the reasons behind the ghosts' deaths while avoiding the ghosts' yeokrin and accurately identifying the truth within a set time limit provide the player with a fun of thrill?	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Play	Play	Can Seo Ji-tae's act of utilizing the 5W1H to uncover the reasons behind the ghosts' deaths while avoiding the ghosts' yeokrin and accurately identifying the truth within a set time limit provide the player with a fun of storytelling?	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Play	Play	Can Seo Ji-tae's act of utilizing the 5W1H to uncover the reasons behind the ghosts' deaths while avoiding the ghosts' yeokrin and accurately identifying the truth within a set time limit affect the final result screen for the player?	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Figure 5: Fourth Iteration's Verification Result

3 Analysis of the Results of Applying the IP-MDA Framework

By establishing a Yeokrin - 5W1H system, we were able to achieve the objectives of the existing game IP while developing an independent system that maintains the sentiment of Han within the original IP.

3.1 IP-MDA Adjustment Results

The implementation of the IP-MDA framework produced significant outcomes through the Yeokrin - 5W1H system, successfully balancing the existing game IP objectives while maintaining the Han sentiment from the original IP. Through a consolidated TMW document, we effectively transformed characters and modernized outdated aspects while staying true to the IP's boundaries. The clear verification criteria enabled us to align aesthetic goals with our established mechanics. The framework's implementation established clear development benchmarks, ensuring unified standards across all elements including art and UI. The verification process facilitated comprehensive discussions about both existing and new mechanics, with the TMW elements and five game IP characteristics guiding developer discussions. This led to practical improvements, such as enhancing difficulty variations through player pressure mechanics. However, we encountered challenges in balancing genre-specific requirements, particularly within the visual novel format's narrative-focused, choice-based gameplay structure.

4 Conclusion

The IP-MDA framework represents a significant advancement in game design methodology, particularly for IP-based game development. Through our case study of the White Day project, we demonstrated that:

The framework successfully bridges the gap between IP characteristics and game mechanics. The iterative verification process ensures alignment with both IP elements and gameplay requirements. Genre considerations play a crucial role in mechanical implementation.

5 Future research directions include

- Expansion of the framework to include genre-specific sub-systems
- Development of standardized verification processes
- Integration of additional transmedia elements

The framework's practical application in the White Day project validates its effectiveness while highlighting areas for potential improvement. As the gaming industry continues to leverage existing IPs, frameworks like IP-MDA will become increasingly valuable for maintaining authenticity while ensuring engaging gameplay.

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